



With prestigious international developers setting up development centres on Indian shores, the scope of opportunities is simply vast"

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artists, game programmers, and for related tasks as the development studios continue to grow rapidly in India," says Tony Garcia, CEO, FXLabs Studios.

It's All About Growth

India is looked upon as a talent-hunter's paradise by companies into animation and gaming. The game development industry also hires animation professionals from that industry. According to a NASSCOM report, *Animation and Gaming India 2007*, about 16,500 animation professionals were employed in the animation industry in 2006. This figure is expected to exceed 26,000 by 2010.

Even so, finding highly-skilled professionals to match global standards still remains a challenge. There are different areas in the gaming industry, such as design, development, testing, and distribution. Game development borrows a lot from the animation industry—the growth of the gaming industry is closely tied with that of the latter.

Game Development: This involves the conception and nurturing of a game, starting from an idea and finally to a working game. "A person seeking a career in game development can pursue a bright career in the different functions, which might be creative and/or technical," says Reena George, general manager, human resources, India Games.

Within game development lie sub-segments such as design, programming, production, visual arts, and audio.

Game Testing: Testing is part of the general game development process, but quite a few companies and studios in India offer testing services exclusively for games to western countries. Those passionate about games and who have good power of observation might want to choose this option. A game tester plays games, observes its working, and finds graphical, audio, and bugs related to the programming. Although the job profile of game testers appears rosy and has lots of fun packed in, they are required to play games for long hours. This might cause physical strain to the eyes, numbness in the fingers, and more!

Business And Distribution: People from the business studies and marketing management domain fit this work profile best. Game distributors promote gaming content. Distribution strategies, dealing with content providers, identifying target audiences, customer feedback and management, etc. fall under this job description. They share information with client companies about gaming habits, tastes, and the preferences of the target audience.

Let's go on to the job functions to be performed at each level within game development.

Loading Stage...

Each function in the industry has levels of hierarchy. For a few of the common designations and sub-segments in game development, refer the table *Teams and Designations*.

Design: A game designer is a jack of all trades. He has a variety of interests and understands the technical side—programming and graphic design. An understanding of psychology and its implementation can be helpful. Knowledge of Artificial Intelligence (AI), too, helps. Tasks here include conceptualisation of a game, design documentation, and working closely with other teams so as to see whether the game is being developed in accordance with the concept. The game design team has a lead designer, who usually evolves by working as a graphic designer or game programmer.

Programming: Graphical elements such as animated characters, game levels, background textures, and sound effects are coded to work together with the game engine. Good coding skills with command over programming languages such as C++, Python, Java, Flash, and other software used in game programming are required.

Being well-versed in programming with OpenGL and DirectX is essential. Knowledge of physics and AI is very beneficial. A programmer should have a balance of technological and creative abilities. "Gaming is about technology, action scripting, animating, etc., but equally, it is about game play, creativity, and an understanding of consumer likes and preferences," informs Carlton D'Silva, creative director, Hungama, who are into the media and entertainment business—viz. gaming portal hungama.com—and also promotions marketing.

Visual Arts: Candidates from the fine arts, animation, and the visual arts / visual communication fields are best suited for the visual arts team. Character and background designing and 2D / 3D modelling and animation are the major activities in this segment. This team works towards constructing the visual elements and the user interface for a game. Students trained on animation software packages also get recruited into this segment. Knowledge of software packages meant for graphics like AutoDesk Maya, 3ds Max, Motion Builder, Adobe Photoshop, Corel Paint, and others makes up the technological side of the equation. For more about animation, turn over to *Life @ 24 fps in Digit*, April 2006.

Teams And Designations

Functional Area	Job Title
Design and Creative Programming	Game designer, level designer, technical designer, script writer Game programmer, AI programmer, network programmer, R&D—engineering programmer, physics programmer, tools programmer, graphics programmer, animation programmer, technical writer, software architect
Visual Arts	Concept artist, texture artist, material specialist, environment artist, storyboard artist, 2D/3D modelling specialist, texture specialist, 2D/3D animator
Production	Game producer, executive game producer, line producer, localisation producer, network game producer, multiplayer producer, advertising producer
Audio	Musician, SFX (sound effects) artist, sound engineer, sound programmer
Business and Other	Sales, marketing, and other top management profiles such as chief executive officer, chief operating officer, vice president— design, programming, production
Testing	Game tester, platform specialist (PC, console, mobile phone, handheld)